INTRODUCTION TO HYPERBOLIC DYNAMICS

JEAN-CHRISTOPHE YOCCOZ
Université de Paris-Sud
Département de Mathématiques
Bâtiment 425
91405 Orsay
France

0. Introduction

A diffeomorphism of a smooth (compact) manifold may exhibit a globally (uniformly) hyperbolic behaviour, like Morse-Smale, Anosov, or Axiom A diffeomorphisms. But, even when the global behaviour is not hyperbolic, it occurs very frequently that the set of points whose orbits are constrained to stay in some appropriate open subset of the manifold is compact and has a hyperbolic structure. Such hyperbolic compact invariant sets then provide a good starting point for a global understanding of the dynamics.

We propose to review the semi-global part of the theory of hyperbolic dynamics, which is concerned with the study of hyperbolic compact invariant sets and basic sets. We leave aside the topics related to global behaviour, such as structural and Ω-stability, Morse-Smale, Anosov, or Axiom A diffeomorphisms. The λ-lemma, the Birkhoff-Smale theorem, and Markov partitions are other important results which are not treated here.

Throughout the theory, it is convenient to work at three successive levels: with Lipschitz perturbations of hyperbolic automorphisms in a Banach space, with Lipschitz perturbations of hyperbolic fibered automorphisms in a finite-dimensional vector bundle (over a compact base), and finally with the actual diffeomorphism in a neighbourhood of a compact invariant hyperbolic subset.

The first part is devoted to the equivalent formulations, spectral and geometrical, of the concept of (uniform) hyperbolicity. We then present the cone field criterion, whose importance, both practical and theoretical, cannot be underestimated.

The second part is concerned with the persistence of hyperbolicity, expansivity, and the shadowing lemma; all these properties ultimately depend on a fixed point theorem for Lipschitz perturbations of hyperbolic automorphisms.

In the third part, we develop the basic properties of stable and unstable manifolds, except for the results related to the transversal regularity of stable and unstable foliations.

Finally, the last part introduces the local product structure, basic sets, and the spectral decomposition of locally maximal chain recurrent sets. The last result is a transition to the global aspects of the theory.

Two appendices are related to the fixed point theorem and elementary topological

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dynamics.

I am heavily indebted to M. Shub ([Sh]) for the treatment of many parts of the theory. The proof of the stable manifold theorem follows Irwin ([I], see also [FHY]).

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1. Compact Invariant Hyperbolic Sets

1.1. Hyperbolic Linear Maps

Let E be a Banach space and $T: E \to E$ be a continuous linear map. Recall that the spectrum $\mathrm{Sp}(T)$ of T is the set of complex numbers λ such that $T_{\mathbb{C}} - \lambda 1_{\mathbb{C}}$ is not an automorphism of $E_{\mathbb{C}}$ (where $E_{\mathbb{C}}$, $T_{\mathbb{C}}$ are the complexifications of E, T). It is a nonempty compact subset of $\mathbb{C}$.

We say that T is *hyperbolic* if

$$\mathrm{Sp}(T) \cap \{ |z| = 1 \} = \emptyset.$$

Then there exist subspaces E_s, E_u of E such that

(1) $E = E_s \oplus E_u$,

(2) $T(E_s) \subset E_s, \quad T(E_u) = E_u$,

(3) $\mathrm{Sp}\left(T|E_s\right) = \mathrm{Sp}(T) \cap \{ |z| < 1 \}$,
$\mathrm{Sp}\left(T|E_u\right) = \mathrm{Sp}(T) \cap \{ |z| > 1 \}$.

We say that E_s (resp. E_u) is the *stable* (resp. *unstable*) subspace of T. If T is hyperbolic, there exist $0 < \kappa_s < 1 < \kappa_u$ such that

(4) $\mathrm{Sp}(T) \cap \{ \kappa_s \leq |z| \leq \kappa_u \} = \emptyset$,

and we say that T is $\kappa_s - \kappa_u$ hyperbolic. Condition (4) is equivalent to the existence of $c > 0$, $\kappa_s' < \kappa_s$, $\kappa_u' > \kappa_u$ such that, for all $n \geq 0$:

(4′) $\left\| \left(T|E_s\right)^n \right\| \leq c\,\kappa_s'^{\,n}, \qquad \left\| \left(T|E_u\right)^{-n} \right\| \leq c\,\kappa_u'^{\,-n}.$

1.2. Hyperbolic Fibered Linear Automorphisms

Let K be a compact metric space and $\pi: E \to K$ a finite-dimensional vector bundle over K. For $x \in K$, we denote by E_x the fiber $\pi^{-1}(x)$.

We consider a homeomorphism f of K and a linear automorphism F of E fibered over f: for each $x \in K$, the restriction F_x of F to E_x is a linear isomorphism from E_x onto $E_{f(x)}$ which depends continuously on x.

We denote by $\Gamma_b(E)$ (resp. $\Gamma_c(E)$) the Banach space of bounded (resp. continuous) sections of π.

For a section σ of π, we define a section $\mathcal{F}(\sigma)$ by:

$$F_x(\sigma(x)) = \mathcal{F}(\sigma)(f(x)).$$

The map $\mathcal{F}$ is linear and leaves $\Gamma_b(E)$, $\Gamma_c(E)$ invariant.

We choose some continuous family $(\| \quad \|_x)_{x\in K}$ of norms on $(E_x)_{x\in K}$. Let $0 < \kappa_s < 1 < \kappa_u$.

Proposition. *The following properties of F are equivalent:*

(i) *$\mathcal{F}$ induces a $\kappa_s - \kappa_u$ hyperbolic automorphism of $\Gamma_b(E)$;*

(i′) *$\mathcal{F}$ induces a $\kappa_s - \kappa_u$ hyperbolic automorphism of $\Gamma_c(E)$;*

(ii) *There exist a direct sum $E = E^s \oplus E^u$ into continuous subbundles and constants $c > 0$, $\kappa'_s < \kappa_s$, $\kappa'_u > \kappa_u$, such that*

a) $F(E^s) = E^s$, $\quad F(E^u) = E^u$;

b) For all $n \geq 0$, $x \in K$, $\left\|F^n|E^s_x\right\| < c{\kappa'_s}^n$, $\quad \left\|F^{-n}|E^u_x\right\| < c{\kappa'_u}^{-n}$.

We then say that F is $\kappa_s - \kappa_u$ hyperbolic, and say that E^s (resp. E^u) is the stable *(resp.* unstable*) subbundle of F.*

Proof. As $\Gamma_c(E)$ is a closed $\mathcal{F}$-invariant subspace of $\Gamma_b(E)$, the spectrum of $\mathcal{F}$ in $\Gamma_c(E)$ is contained in the spectrum of $\mathcal{F}$ in $\Gamma_b(E)$. Hence (i) $\Rightarrow$ (i′).

If (ii) holds, the splitting

$$\Gamma_b(E) = \Gamma_b(E^s) \oplus \Gamma_b(E^u)$$

satisfies conditions (1), (2), (3), (4′) of 1.1. Therefore (ii) $\Rightarrow$ (i).

To prove (i′) $\Rightarrow$ (ii), we use the following standard

Lemma. *Let $\mathcal{T} : \Gamma_c(E) \to \Gamma_c(E)$ be a $C(K)$-linear continuous map. Then there exists a continuous family $(T_x)_{x\in K} \in (L(E_x))_{x\in K}$ such that $\mathcal{T}\sigma(x) = T_x(\sigma(x))$ for all $x \in K$, $\sigma \in \Gamma_c(E)$.*

Proof of lemma.

a) If $\sigma \in \Gamma_c(E)$ vanishes on an open set U, $\mathcal{T}\sigma$ vanishes on U: for any $x \in U$, let $\rho \in C(K)$ such that $\rho(x) = 1$, $\rho\sigma \equiv 0$; we have $\mathcal{T}\sigma(x) = \rho(x)\mathcal{T}\sigma(x) = \mathcal{T}(\rho\sigma)(x) = 0$.

b) If $\sigma \in \Gamma_c(E)$ vanishes at a point $x \in K$, $\mathcal{T}\sigma(x) = 0$: indeed there exists a sequence $(\sigma_n)_{n\geq 0}$ with $\lim \sigma_n = \sigma$ and $\sigma_n(y) = 0$ for $d(x,y) < 2^{-n}$. Then

$$\mathcal{T}\sigma(x) = \lim_n \mathcal{T}\sigma_n(x) = 0$$

by a).

c) Let $x \in K$, $v \in E_x$; let $\sigma \in \Gamma_c(E)$ such that $\sigma(x) = v$; by b), the vector $\mathcal{T}\sigma(x)$ does not depend on the choice of σ, but only on x and v. Call it $T_x(v)$. The family $(T_x)_{x\in k}$ is readily seen to satisfy the conclusion of the lemma. □

Proof of (i′) $\Rightarrow$ (ii). Let $\Gamma_c(E) = \mathcal{E}_s \oplus \mathcal{E}_u$ be the splitting associated to the linear hyperbolic automorphism $\mathcal{F}$ of $\Gamma_c(E)$; let π^s, π^u be the corresponding projections on $\mathcal{E}_s, \mathcal{E}_u$. For $\sigma \in \Gamma_c(E)$, $\rho \in C(K)$, we have

$$\mathcal{F}(\rho\sigma) = \rho \circ f^{-1}\mathcal{F}(\sigma),$$

hence, for all $n \geq 0$:

$$\mathcal{F}^n(\rho\pi^s(\sigma)) = \rho \circ f^{-n}\mathcal{F}^n(\pi^s(\sigma)),$$
$$\mathcal{F}^{-n}(\rho\pi^u(\sigma)) = \rho \circ f^{n}\mathcal{F}^{-n}(\pi^u(\sigma)).$$

We conclude that $\rho\pi^s(\sigma) \in \mathcal{E}_s$, $\rho\pi^u(\sigma) \in \mathcal{E}_u$. Therefore π^s and π^u are $C(K)$-linear. Let $(p_x^s)_{x\in K}$, $(p_x^u)_{x\in K}$ be the families associated to π^s, π^u by the lemma. From $(\pi^s)^2 = \pi^s$, $(\pi^u)^2 = \pi^u$, $\pi^s + \pi^u = 1$ we deduce for any $x \in K$ the same relations for p_x^s, p_x^u. Let $E_x^s = p_x^s(E_x)$, $E_x^u = p_x^u(E_x)$. As p_x^s, p_x^u depend continuously on x, the families $(E_x^s)_{x\in K}$, $(E_x^u)_{x\in K}$ form continuous subbundles E^s, E^u of E which satisfy property (ii). □

1.3. Hyperbolic Compact Invariant Sets

Let M be a smooth manifold and $f: M \to M$ be a C^1-diffeomorphism.

A subset K of f is *invariant* if $f(K) = K$.

A compact invariant subset K of M is *hyperbolic* if the restriction of the tangent map Tf to $TM_{|K}$ is *hyperbolic* in the sense of 1.2.

1.4 Adapted Norms

1.4.1.

Let $(E, \|\ \ \|)$ be a Banach space, $0 < \kappa_s < 1 < \kappa_u$, and T a $\kappa_s - \kappa_u$ hyperbolic linear endomorphism of E with associated splitting $E = E_s \oplus E_u$.

We say that a norm $|\ \ |$ on E is *adapted* to T if it is equivalent to $\|\ \ \|$ and satisfies

(i) $|x_s + x_u| = \max(|x_s|, |x_u|), \quad x_s \in E_s,\ x_u \in E_u;$

(ii) $\left|T|E_s\right| \leq \kappa_s, \qquad \left|\left(T|E_u\right)^{-1}\right| \leq \kappa_u^{-1}$

Such a norm always exists; one can for instance take, with N big enough:

$$|x_s| = \sum_{i=0}^{N-1} \kappa_s^{-i} \left\|T^i(x_s)\right\|, \quad x_s \in E_s,$$
$$|x_u| = \sum_{i=0}^{N-1} \kappa_u^{i} \left\|T^{-i}(x_u)\right\|, \quad x_u \in E_u,$$

and impose (i).

1.4.2.

In the same spirit, if F is a linear fibered automorphism which is $\kappa_s - \kappa_u$ hyperbolic as in 1.2., one can find a continuous family of norms $(|\ \ |_x)_{x\in K}$ on $(E_x)_{x\in K}$ which is *adapted* to F in the following sense:

(i) $|v_s + v_u|_x = \max(|v_s|_x, |v_u|_x), \quad$ for $x \in K$, $v_s \in E_x^s$, $v_u \in E_x^u$;

(ii) $\left|F_x|E_x^s\right| \leq \kappa_s, \qquad \left|\left(F_x|E_x^u\right)^{-1}\right| \leq \kappa_u^{-1}.$

1.5. The Cone Field Criterion

Cone fields are the basic tool to prove that some domain in a manifold contains a compact invariant hyperbolic subset for a diffeomorphism.

Let $\pi: E \to K$ be a finite-dimensional vector bundle over a compact metric space K. For each $x \in K$, let $E_x = E_x^1 \oplus E_x^2$ be a splitting (not assumed to be continuous in x), and let $|\ \ |_{1,x}$, $|\ \ |_{2,x}$ be norms on E_x^1, E_x^2 respectively.

Define

$$C_x = \{v = v_1 + v_2,\ v_1 \in E_x^1, v_2 \in E_x^2, |v_1|_{1,x} \leq |v_2|_{2,x}\}.$$

The family $(C_x)_{x\in K}$ is the *cone field* in E associated to the given data.

The dual cone field is the family $(C_x^*)_{x\in K}$ defined by $C_x^* = E_x - \operatorname{int} C_x$.

The cone field $(C_x)_{x\in K}$ is *bounded* if, given some (and then any) continuous family of norms $(\|\ \ \|_x)_{x\in K}$ on E, there exists $c > 0$ such that

$$c^{-1} \|v_1 + v_2\|_x \leq \max\left(|v_1|_{1,x}, |v_2|_{2,x}\right) \leq c \|v_1 + v_2\|_x,$$

for $x \in K$, $v_1 \in E_x^1$, $v_2 \in E_x^2$. This amounts to saying that $\Gamma_b(E) = \Gamma_b(E^1) \oplus \Gamma_b(E^2)$ as Banach spaces.

Now let F be a linear automorphism of E fibered over a homeomorphism f of K. Assume that there exist a bounded cone field $(C_x)_{x\in K}$ as above, constants $\lambda, \mu > 1$ and an integer $m \geq 1$ such that:

(i) for $i = 1, 2$, $x \in K$, the dimension d_x^i of E_x^i is equal to $d_{f(x)}^i$;

(ii) for $x \in K$, $F(C_x) \subset \widetilde{C}_{f(x)}$, where

$$\widetilde{C}_x = \{v = v_1 + v_2 \in E_x, \lambda |v_1|_{1,x} \leq |v_2|_{2,x}\};$$

(iii) for $x \in K, v \in C_x$,

$$\|F^m(v)\|_{f^m(x)} \geq \mu \|v\|_x;$$

(iii′) for $x \in K, v \in C_x^*$,

$$\|F^{-m}(v)\|_{f^{-m}(x)} \geq \mu \|v\|_x.$$

Then, one can conclude that F is hyperbolic.

To see this, it is sufficient to find for each $x \in K$ subspaces E_x^s and E_x^u which satisfy

a) $\dim E_x^s = d_x^1$, $\dim E_x^u = d_x^2$;
b) $E_x^s \subset C_x^*$, $E_x^u \subset C_x$;
c) $F_x(E_x^s) = E_{f(x)}^s$, $F_x(E_x^u) = E_{f(x)}^u$.

Indeed, we will then have from (ii) that $E_x^u \subset \widetilde{C}_x$; from this and the boundedness of the cone field, we get an invariant splitting $\Gamma_b(E) = \Gamma_b(E^s) \oplus \Gamma_b(E^u)$ which allows us (using (iii), (iii′)) to check property (i) of proposition 1.2.

Let $(f^n(x))_{n\in\mathbb{Z}}$ be an orbit of f in K. We distinguish two cases:

1) x is not periodic: For $y \in K$, let $G(y)$ be the compact set whose elements are the d_y^2-planes contained in C_y; F_y induces a continuous map $\widetilde{F}_y$ from $G(y)$ to $G(f(y))$. The intersection

$$G = \bigcap_{n \geq 0} \widetilde{F}^n \left(G\left(f^{-n}x\right)\right)$$

is nonempty. Choose $E_x^u \in G$ and define $E_{f^n(x)}^u = F^n(E_x^u)$ for $n \in \mathbb{Z}$.

2) x is periodic: Let k be the period of x, $A = F^k|_{E_x}$. If A had a (complex) eigenvalue of modulus 1, there would exist $v \neq 0$ in E_x and $c > 0$ such that

$$c^{-1} \|v\|_x \leq \|A^n v\|_x \leq c \|v\|_x, \qquad \forall\, n \in \mathbb{Z}.$$

This is impossible: $v \in C_x$ would contradict (iii), and $v \in C_x^*$ would contradict (iii′). Hence, A is hyperbolic. Take for E_x^s (resp. E_x^u) the stable (resp. unstable) subspace of A; then by (iii), (iii′) we have $E_x^s \cap C_x = E_x^u \cap C_x^* = \{0\}$, which implies a) and b). Define $E_{f^i(x)}^s = F^i(E_x^s)$ and $E_{f^i(x)}^u = F^i(E_x^u)$ for $1 \leq i < k$. The proof of the hyperbolicity of F is complete.

Proposition. *Let f be a C^1-diffeomorphism of a smooth manifold M, and K a compact invariant hyperbolic subset. There exist a neighbourhood $\mathcal{U}$ of f in* $\mathrm{Diff}^1(M)$ *and a compact neighbourhood U of K in M, such that, for any $g \in \mathcal{U}$, the compact g-invariant set*

$$\bigcap_{n \in \mathbb{Z}} g^{-n}(U)$$

is hyperbolic.

Proof. Choose a continuous family $(|\ \ |_x)_{x \in K}$ of norms on $E = TM|_K$ which is adapted to $Tf|_K$. As the dimension of E_x^s is locally constant on K, we can assume that it takes a constant value d^s. Using a partition of unity, we can construct for y in a neighbourhood V of K a d^s-plane $E_y^1 \subset T_yM$ depending continuously on y such that $E_x^1 = E_x^s$ for $x \in K$. Similarly, we get a d^u-plane E_y^2 such that $E_y^2 = E_y^u$ for $y \in K$. Extend continuously the family $(|\ \ |_x)_{x \in K}$ to V.

Let $(C_y)_{y \in K}$ be the bounded cone field defined by these data (restricting V if necessary). By continuity, there exist a neighbourhood $\mathcal{U}$ of f in $\mathrm{Diff}^1(M)$ and a compact neighbourhood $U \subset V$ of K such that the conditions (i), (ii), (iii), (iii′) of the cone field criterion are satisfied (with appropriate λ, μ, m) for $g \in \mathcal{U}$ on $\bigcap_{n \in \mathbb{Z}} g^{-n}(U)$. $\square$

2. Hyperbolic Continuation

2.1. A Fixed Point Theorem

Let E be a Banach space and T be a linear hyperbolic endomorphism of E. Let $\kappa_s < 1 < \kappa_u$ such that T is $\kappa_s - \kappa_u$ hyperbolic; equip E with a norm adapted to T.

Theorem. *Let $f: E \to E$ be a Lipschitz map such that:*

$$\varepsilon = \operatorname{Lip}(f - T) < \varepsilon_0 = \min\left(1 - \kappa_s, 1 - \kappa_u^{-1}\right)$$

Then f has a unique fixed point p in E and we have

$$|p| < (\varepsilon_0 - \varepsilon)^{-1} |f(0)|.$$

Proof. Identify E with $E_s \times E_u$; let $p_s : E \to E_s$, $p_u : E \to E_u$ be the canonical projections, $f_s = p_s \circ f$, $f_u = p_u \circ f$. Define:

$$\overline{f}(x_s, x_u) = \left(f_s(x_s, x_u), x_u + T_u^{-1}(x_u - f_u(x_s, x_u))\right), \qquad x_s \in E_s, \quad x_u \in E_u$$

where $T_s = T|E_s$, $T_u = T|E_u$.

Clearly f and $\overline{f}$ have the same fixed points.

Let $x_s, x_s' \in E_s$, $x_u, x_u' \in E_u$. We have (with $x = (x_s, x_u), x' = (x_s', x_u')$):

$$\begin{aligned}
&|(f_s(x_s, x_u) - T_s(x_s)) - (f_s(x_s', x_u') - T_s(x_s'))| \le \varepsilon |x - x'|, \\
&|(f_u(x_s, x_u) - T_u(x_u)) - (f_u(x_s', x_u') - T_u(x_u'))| \le \varepsilon |x - x'|, \\
&|T_s| \le \kappa_s, \qquad \left|T_u^{-1}\right| \le \kappa_u^{-1}
\end{aligned}$$

hence

$$\begin{aligned}
&\left|p_s \circ \overline{f}(x_s, x_u) - p_s \circ \overline{f}(x_s', x_u')\right| \le (\kappa_s + \varepsilon)|x - x'|, \\
&\left|p_u \circ \overline{f}(x_s, x_u) - p_u \circ \overline{f}(x_s', x_u')\right| \le \kappa_u^{-1}(1 + \varepsilon)|x - x'|,
\end{aligned}$$

and finally

$$\left|\overline{f}(x_s', x_u') - \overline{f}(x_s, x_u)\right| \le (1 + \varepsilon - \varepsilon_0)|x - x'|.$$

Hence $\overline{f}$ is contracting; as $\left|\overline{f}(0)\right| \le |f(0)|$, the theorem follows. □

2.2. The Grobman-Hartman Theorem

Let E be a Banach space and T be a linear hyperbolic *automorphism* of E. Choose $\kappa_s < 1 < \kappa_u$ such that T is $\kappa_s - \kappa_u$ hyperbolic and a norm on E adapted to T.

Let $f : E \to E$ be a map such that $\Delta f = f - T$ is Lipschitz. If $\operatorname{Lip} \Delta f < |T^{-1}|^{-1}$, f is a homeomorphism of E: indeed, for $x, y \in E$ we have

$$y = f(x) \iff x = T^{-1}(y - \Delta f(x)) = \theta(x, y),$$

with θ continuous and uniformly contracting in x.

Theorem. *Assume that Δf is* bounded *and that*

$$\varepsilon = \operatorname{Lip} \Delta f < \varepsilon_1 = \min\left(\left|T^{-1}\right|^{-1}, 1 - \kappa_s, 1 - \kappa_u^{-1}\right).$$

Then there exists a unique homeomorphism $h = id_E + \Delta h$ of E such that Δh is bounded and

$$h \circ T \circ h^{-1} = f$$

Proof. Let $f = T + \Delta f$, $g = T + \Delta g$ such that Δf, Δg are bounded, and Lipschitz, with $\mathrm{Lip}\,(\Delta f)$, $\mathrm{Lip}\,(\Delta g) < \varepsilon_1$.

We first show that there exists a unique continuous bounded map $\Delta h\colon E \to E$ such that

$$f \circ (id_E + \Delta h) = (id_E + \Delta h) \circ g. \tag{1}$$

As g is an homeomorphism, this is equivalent to:

$$(T + \Delta f) \circ (id_E + \Delta h) \circ g^{-1} = id_E + \Delta h$$

or

$$T \circ \Delta h \circ g^{-1} + \Delta f \circ (id_E + \Delta h) \circ g^{-1} + T \circ g^{-1} - id_E = \Delta h. \tag{1'}$$

Let $\mathcal{E} = C_b\,(E, E)$ be the space of bounded continuous maps, equipped with the sup norm. Let $\mathcal{E}_s = C_b\,(E, E_s)$, $\mathcal{E}_u = C_b\,(E, E_u)$, we have

$$\mathcal{E} = \mathcal{E}_s \oplus \mathcal{E}_u.$$

For $\Delta h \in \mathcal{E}$, define

$$\begin{aligned} \mathcal{T}\,(\Delta h) &= T \circ \Delta h \circ g^{-1} \in \mathcal{E}, \\ \Delta\mathcal{F}\,(\Delta h) &= \Delta f \circ (id_E + \Delta h) \circ g^{-1} + T \circ g^{-1} - id_E \in \mathcal{E}, \end{aligned}$$

(as $T - g$ is bounded, $T \circ g^{-1} - id_E$ is bounded).

Clearly, $\mathcal{T}$ is hyperbolic, and more precisely:

$$\begin{aligned} \mathcal{T}\,(\mathcal{E}_s) &= \mathcal{E}_s, \quad \left|\mathcal{T}_{|\mathcal{E}_s}\right| \le \kappa_s, \\ \mathcal{T}\,(\mathcal{E}_u) &= \mathcal{E}_u, \quad \left|\left(\mathcal{T}_{|\mathcal{E}_u}\right)^{-1}\right| \le \kappa_u^{-1}. \end{aligned}$$

Also, $\Delta\mathcal{F}$ is Lipschitz with

$$\mathrm{Lip}\,\Delta\mathcal{F} \le \mathrm{Lip}\,\Delta f$$

From Theorem 2.1, we see that $\mathcal{T} + \Delta\mathcal{F}$ has a unique fixed point in $\mathcal{E}$, hence (1) (or (1′)) has a unique solution in $\mathcal{E}$.

To conclude the proof of the theorem, it is sufficient to see that the map $id + \Delta h$ satisfying (1) has to be a *homeomorphism*. Indeed, there exists $\overline{\Delta h} \in \mathcal{E}$ such that:

$$g \circ \left(id_E + \overline{\Delta h}\right) = \left(id_E + \overline{\Delta h}\right) \circ f \tag{2}$$

Then we have

$$\begin{aligned} g \circ \left(id_E + \overline{\Delta h}\right) \circ (id_E + \Delta h) &= \left(id_E + \overline{\Delta h}\right) \circ (id_E + \Delta h) \circ g \\ f \circ (id_E + \Delta h) \circ \left(id_E + \overline{\Delta h}\right) &= (id_E + \Delta h) \circ \left(id_E + \overline{\Delta h}\right) \circ f \end{aligned}$$

These equations being of the same type as (1), uniqueness of solutions show that

$$\left(id_E + \overline{\Delta h}\right) \circ (id_E + \Delta h) = (id_E + \Delta h) \circ \left(id_E + \overline{\Delta h}\right) = id_E. \quad \square$$

2.3. An Invariant Section Theorem

Let $\pi : E \to K$ be a finite-dimensional vector bundle over a compact metric space K, and $F : E \to E$ a linear automorphism fibered over a homeomorphism f of K. We assume that F is hyperbolic, choose $\kappa_s < 1 < \kappa_u$ such that F is $\kappa_s - \kappa_u$ hyperbolic and a continuous family $(|\quad|_x)_{x\in K}$ of norms adapted to F.

Consider a continuous map $\widetilde{F}: E \to E$ fibered over f.

Theorem. *Suppose that for each $x \in K$ the restriction $\widetilde{F}_x: E_x \to E_{f(x)}$ is Lipschitz and that*

$$\varepsilon = \sup_K \left(\mathrm{Lip}\left(\widetilde{F}_x - F_x\right)\right) < \min\left(1-\kappa_s, 1-\kappa_u^{-1}\right) = \varepsilon_0$$

Then there exists a unique bounded *section σ such that $\mathcal{F}(\sigma) = \sigma$. Moreover, σ is* continuous *and we have*

$$|\sigma| < (\varepsilon_0 - \varepsilon)^{-1} \sup_K \left|\widetilde{F}_x(0_x)\right|.$$

Proof. The formula

$$\widetilde{\mathcal{F}}\sigma(f(x)) = \widetilde{F}_x(\sigma(x))$$

defines a map $\widetilde{\mathcal{F}}: \Gamma_b(E) \to \Gamma_b(E)$, leaving $\Gamma_c(E)$ invariant. The hypotheses of Theorem 2.1 are satisfied by $\widetilde{\mathcal{F}}$ and $\widetilde{\mathcal{F}}|\Gamma_c(E)$; the conclusions of this theorem readily imply the conclusions of the theorem above. □

2.4. A Technical Device

Let M be a smooth manifold. We denote by Exp a smooth diffeomorphism from an open neighbourhood of the zero section in TM onto an open neighbourhood of the diagonal in $M \times M$, of the form

$$\mathrm{Exp}(v) = (x, \exp_x(v)), \quad x \in M, v \in T_xM,$$

where $\exp_x$ is a smooth diffeomorphism from a neighbourhood of 0_x in T_xM onto a neighbourhood of x in M, with $\exp_x(0_x) = x$. We can use for instance the exponential map associated to some Riemannian metric on M.

Let f be a C^1-diffeomorphism of M, and K a compact invariant subset of M. We equip $E = TM_{|K}$ with a continuous family $(\|\quad\|_x)_{x\in K}$ of Euclidean norms.

There exist δ_0 and a neighbourhood $\mathcal{U}$ of f in $\mathrm{Diff}^1(M)$, such that for $g \in \mathcal{U}$, $x \in K$, $v \in T_xM$, we have that $g(\exp_x(v))$ belongs to the image of $\exp_{f(x)}$ if $\|v\|_x \le 2\delta_0$. Define then

$$\overline{G}_x(v) = \exp_{f(x)}^{-1} \circ g \circ \exp_x(v).$$

On $\{\|v\|_x \le 2\delta_0\}$, the map $\overline{G}_x$ is C^1-close to T_xf if $\delta_0, \mathcal{U}$ are small enough.

Fix a smooth function $\eta : \mathbb{R} \to \mathbb{R}$ such that $\eta(t) = 0$ for $t \ge 2$, $\eta(t) = 1$ for $t \le 1$. With $\delta < \delta_0$, define

$$\widetilde{G}_x(v) = T_xf(v) + \eta\left(\delta^{-1}\|v\|\right)\left(\overline{G}_x(v) - T_xf(v)\right).$$

This makes sense for all $v \in T_x M$. The family $(\widetilde{G}_x)_{x \in K}$ defines a continuous map $\widetilde{G} : E \to E$ fibered over f. One has, for $x \in K$, $v \in E_x$:

$$\begin{aligned} \widetilde{G}(v) &= T_x f(v), && \text{if } \|v\| \geq 2\delta, \\ \exp_{f(x)} \widetilde{G}(v) &= g(\exp_x(v)), && \text{if } \|v\| \leq \delta, \end{aligned} \tag{1}$$

By taking δ, $\mathcal{U}$ small enough, we can make $\widetilde{G}_x$ arbitrarily uniformly C^1-close to $T_x f$.

2.5. The Expansivity Property

Let M be a smooth manifold, d a distance on M defining its topology, f a C^1-diffeomorphism of M, and K a compact invariant hyperbolic subset. We write $E = TM_{|K}$.

Proposition. *There exist $\varepsilon > 0$ and a neighbourhood U of K in M such that if x, y are two distinct points in $\bigcap_{n \in \mathbb{Z}} f^{-n}(U)$, we have*

$$\sup_{n \in \mathbb{Z}} d(f^n(x), f^n(y)) \geq \varepsilon.$$

Proof. Choose a compact neighbourhood U such that $\bigcap_{n \in \mathbb{Z}} f^{-n}(U) = K_1$ is hyperbolic (Proposition 1.5). For $g = f$, K_1 and δ small enough, construct $\widetilde{F}$ as in 2.4.

The map $\widetilde{F}$ induces a map $\widetilde{\mathcal{F}} : \Gamma_b(E) \to \Gamma_b(E)$ Lipschitz close to the map $\mathcal{F}$ induced by $Tf_{|K}$, and $\widetilde{\mathcal{F}} - \mathcal{F}$ is bounded. Let $x \in K_1$, $v \in E_x$, $v \neq 0$; by the Grobman-Hartman theorem, the orbit under $\widetilde{\mathcal{F}}$ of the section σ_v defined by $\sigma_v(x') = 0$ if $x' \neq x$, $\sigma_v(x) = v$, is unbounded. This means that the orbit of $y = \exp_x v$ under f cannot stay close to the orbit of x (see relation (1) of 2.4). □

2.6. Hyperbolic Continuation

Let f be a C^1-diffeomorphism of a smooth manifold M, and K be a hyperbolic compact invariant subset of M.

Theorem. *There exist a neighbourhood $\mathcal{U}$ of f in $\mathrm{Diff}^1(M)$ and a continuous map $H : \mathcal{U} \to C(K, M)$ with the following properties*

(i) *for any $g \in \mathcal{U}$, $H(g)$ is a continuous embedding of K in M;*

(ii) *$H(f)$ is the canonical inclusion i of K in M;*

(iii) *There exists a neighbourhood $\mathcal{V}$ of i in $C(K, M)$ such that, for any $g \in \mathcal{U}$, $H(g)$ is the unique map in $\mathcal{V}$ satisfying*

$$g \circ H(g) = H(g) \circ f_{|K} \tag{1}$$

(iv) *for any $g \in \mathcal{U}$, $K_g = H(g)(K)$ is a hyperbolic compact invariant subset of M relative to g.*

Proof. With $\mathcal{U}$, δ small enough and $g \in \mathcal{U}$, construct $\widetilde{G}$ as in 2.4.

If $\mathcal{U}$, δ are small enough, it follows from Theorem 2.3 that there exists a unique bounded section σ_g of $TM_{|K}$ invariant under $\widetilde{G}$, and this section is continuous. Restricting $\mathcal{U}$ if necessary, we have $\|\sigma_g(x)\|_x \leq \delta$ for $x \in K$, $g \in \mathcal{U}$. If we define

$$H(g)(x) = \exp_x(\sigma_g(x)), \qquad x \in K,$$

we obtain property (iii). Property (ii) is clear, and property (iv) follows from Proposition 1.5.

The fixed point of f in Theorem 2.1 depends continuously on f, and the map $g \mapsto \widetilde{G}$ is continuous (in the C^1-topology), hence the map $g \mapsto H(g)$ is continuous.

Finally, $H(g)$ is injective (for $\mathcal{U}$ small enough); indeed, let d be a distance on M defining its topology and let $x, x' \in K$ such that $H(g)(x) = H(g)(x')$. If we have

$$d(y, H(g)(y)) < \varepsilon_0, \quad y \in K,$$

it follows for all $n \in \mathbb{Z}$ that

$$\begin{aligned} d(f^n(x), f^n(x')) &\leq d(f^n(x), H(g)(f^n(x))) + d(f^n(x'), H(g)(f^n(x'))) \\ &< 2\varepsilon_0; \end{aligned}$$

from Proposition 2.5 we conclude that $x = x'$ for $2\varepsilon_0 < \varepsilon$. □

Remark. The analogy with analytic continuation may be stronger if we reformulate the last theorem in the following way.

Let L be a compact metric space and $h: L \to L$ a homeomorphism; in the product $C(L, M) \times \mathrm{Diff}^1(M)$, consider the set $\mathcal{D}$ of pairs (i, f) such that $i(f)$ is a hyperbolic (compact) invariant subset for f, $i \circ h = f \circ i$, and i is an embedding.

Then the theorem asserts that $\mathcal{D}$ is a "Riemann domain" over $\mathrm{Diff}^1(M)$ in the sense that the projection from $\mathcal{D}$ to $\mathrm{Diff}^1(M)$ is a *local homeomorphism*.

2.7. The Shadowing Lemma

Let M be a smooth manifold, d a distance on M associated to some Riemannian metric, f a C^1-diffeomorphism of M, and K a compact invariant hyperbolic subset.

Recall that for $\delta > 0$ a δ-pseudo-orbit is a sequence $(x_i)_{i\in\mathbb{Z}}$ such that the distance $d(x_{i+1}, f(x_i)) < \delta$ for all $i \in \mathbb{Z}$.

Theorem (Shadowing Lemma). *There exist a neighbourhood U of K and a constant $\theta > 0$ with the following property: for any $\delta > 0$, and any δ-pseudo-orbit $(x_i)_{i\in\mathbb{Z}}$ contained in U, there exists $x \in M$ such that*

$$d(f^i(x), x_i) \leq \theta\delta, \qquad \textit{for all } i \in \mathbb{Z}.$$

Remark. If δ, U are small enough, the point x is unique by the expansivity property. Moreover, by Proposition 1.5, x belongs to a compact invariant hyperbolic subset $L = \bigcap_{n\in\mathbb{Z}} f^{-n}(V)$, for some appropriate compact neighbourhood $V \supset U$.

Proof. It is sufficient to consider the case where $\delta > 0$ is small. Let $(x_i)_{i\in\mathbb{Z}}$ be a δ-pseudo-orbit lying in U, with U small enough.

Choose, for $i \in \mathbb{Z}$, a point $\overline{x}_i \in K$ among the closest to x_i. Then $f(\overline{x}_i)$ is close to $\overline{x}_{i+1}$. Let $E_i = T_{\overline{x}_i} M$, $E_i^s = E_{\overline{x}_i}^s$, $E_i^u = E_{\overline{x}_i}^u$, and let a_i, $v_i \in E_i$ such that

$$\exp_{\overline{x}_i}(a_i) = f(\overline{x}_{i-1}),$$
$$\exp_{\overline{x}_i}(v_i) = x_i.$$

Consider $T_{\overline{x}_i} f$ as an affine map from E_i to E_{i+1} sending 0_{E_i} to a_{i+1}, and choose a linear map $T_i : E_i \to E_{i+1}$ close to $T_{\overline{x}_i} f$ such that

$$T_i(E_i^s) = E_{i+1}^s, \quad T_i(E_i^u) = E_{i+1}^u.$$

Let $\mathcal{E} = \{ (w_i)_{i\in\mathbb{Z}}, \quad w_i \in E_i, \quad \sup_i \|w_i\| = |||(w_i)_{i\in\mathbb{Z}}||| < +\infty \}$.

The family $(T_i)_{i\in\mathbb{Z}}$ defines a linear automorphism T of $\mathcal{E}$ which is hyperbolic if δ, U are small enough.

Construct as in 2.4 maps $\widetilde{F}_i : E_i \to E_{i+1}$ such that, for some $\delta_1 > 0$, we have

$$\exp_{\overline{x}_{i+1}} \widetilde{F}_i(w) = f\left(\exp_{\overline{x}_i} w\right), \quad \|w\| \le \delta_1,$$

and moreover $\operatorname{Lip}(\widetilde{F}_i - T_i)$ is uniformly small. With δ, U small enough, we have $\|v_i\| \le \frac{1}{2}\delta_1$ for all $i \in \mathbb{Z}$ and

$$\|\widetilde{F}_i(v_i) - v_{i+1}\| \le \theta_1 \delta,$$

for some constant $\theta_1 > 0$.

It now follows from Theorem 2.1 that if δ, U are small enough there exists a unique sequence $(w_i)_{i\in\mathbb{Z}}$ in $\mathcal{E}$ such that $\widetilde{F}_i(w_i) = w_{i+1}$. Moreover, we have

$$\|v_i - w_i\| \le \theta_2 \delta,$$

by Theorem 2.1. With δ small enough, this implies $\|w_i\| \le \delta_1$; if we define $x = \exp_{\overline{x}_0}(w_0)$, we will also have from the inequality above that

$$d\left(f^i(x), x_i\right) \le \theta\delta,$$

because $x_i = \exp_{\overline{x}_i}(v_i)$, $f^i(x) = \exp_{\overline{x}_i}(w_i)$. $\square$

Recall that a point x is chain recurrent (for f) if for any $\delta > 0$ there exists a periodic δ-pseudo-orbit through x.

Corollary. *Let M, f, K, be as above and $x_0 \in K$ a chain recurrent point for $f|_K$. Then x_0 is accumulated by hyperbolic periodic points whose orbits are arbitrarily close to K.*

This follows immediately from the remark after the theorem: if $(x_i)_{i\in\mathbb{Z}}$ is a periodic δ-pseudo-orbit of period k, and x satisfies

$$d\left(f^i(x), x_i\right) \le \theta\delta,$$

one deduces from the uniqueness of x (for δ small enough) that $f^k(x) = x$.

3. Stable and Unstable Manifolds

3.1. The Stable Manifold Theorem

Let E be a Banach space and T be a hyperbolic linear endomorphism of E. We choose $0 < \kappa_s < 1 < \kappa_u$ such that T is $\kappa_s - \kappa_u$ hyperbolic, and equip E with a norm adapted to T. We can therefore identify E with $E_s \times E_u$.

Let $\kappa \in (\kappa_s, \kappa_u)$; for any map $f: E \to E$, define

$$W_\kappa^s(f) = \{x \in E, \sup_{n \geq 0} \kappa^{-n} |f^n(x)| < +\infty\}.$$

Then we have $W_\kappa^s(T) = E_s$, and more precisely:

$$\lim_{n \to +\infty} \kappa^{-n} |T^n(x)| = 0, \qquad \forall x \in E^s.$$

Observe also that one always has $f(W_\kappa^s(f)) \subset W_\kappa^s(f)$.

Theorem. *Let $f: E \to E$ be a Lipschitz map such that $f(0) = 0$ and*

$$\varepsilon = \mathrm{Lip}(f - T) < \varepsilon_2 = \min(\kappa - \kappa_s, \kappa_u - \kappa)$$

Then $W_\kappa^s(f)$ is the graph of a contracting map $g: E_s \to E_u$, satisfying $g(0) = 0$ and we have

$$\mathrm{Lip}\left(f|W_\kappa^s(f)\right) \leq \varepsilon + \kappa_s,$$

$$\lim_{n \to +\infty} \kappa^{-n} |f^n(x)| = 0, \qquad \forall x \in W_\kappa^s(f).$$

Remark. The theorem actually holds, with the same proof, whenever $0 < \kappa_s < \kappa_u$ and

$$\mathrm{Sp}(T) \cap \{\kappa_s \leq |z| \leq \kappa_u\} = \emptyset.$$

The hypothesis $\kappa_s < 1 < \kappa_u$ is here irrelevant.

3.2. Proof of the Theorem

Let $T_s = T|E_s$, $T_u = T|E_u$, and $p_s: E \to E_s$, $p_u: E \to E_u$ be the canonical projections. Let $f_s = p_s \circ f$, $f_u = p_u \circ f$.

To any sequence $(x^n)_{n \geq 0} = (x_s^n, x_u^n)$ in E, we associate sequences $(y_s^n)_{n \geq 1}$ in E_s and $(y_u^n)_{n \geq 0}$ in E_u defined by:

$$\begin{cases} y_s^{n+1} = f_s(x_s^n, x_u^n) \\ y_u^n = x_u^n + T_u^{-1}\left(x_u^{n+1} - f_u(x_s^n, x_u^n)\right) \end{cases} \qquad \forall n \geq 0 \quad \text{(cf. 2.1)}$$

If $(\overline{x}^n)_{n \geq 0}$ is another sequence in E, with associated sequences $(\overline{y}_s^n)_{n \geq 1}$, $(\overline{y}_u^n)_{n \geq 0}$, we have for $n \geq 0$ (cf. 2.1):

$$\left|y_s^{n+1} - \overline{y}_s^{n+1}\right| \leq (\kappa_s + \varepsilon)\left|x^n - \overline{x}^n\right|, \tag{1}$$

$$|y_u^n - \overline{y}_u^n| \leq \kappa_u^{-1} \left(\left| x^{n+1} - \overline{x}^{n+1} \right| + \varepsilon \left| x^n - \overline{x}^n \right| \right). \tag{2}$$

Observe also that, for $n \geq 0$:

$$f(x^n) = x^{n+1} \Longleftrightarrow \begin{cases} x_s^{n+1} = y_s^{n+1} \\ x_u^n = y_u^n. \end{cases} \tag{3}$$

Let

$$\mathcal{E}_s = \{(x_s^n)_{n\geq 1}, x_s^n \in E_s, \|(x_s^n)\| = \sup_n \kappa^{-n} |x_s^n| < +\infty\}$$
$$\mathcal{E}_u = \{(x_u^n)_{n\geq 0}, x_u^n \in E_u, \|(x_u^n)\| = \sup_n \kappa^{-n} |x_u^n| < +\infty\}.$$

Equip $\mathcal{E}_s \times \mathcal{E}_u$ and $E_s \times \mathcal{E}_s \times \mathcal{E}_u$ with the sup norms. Let $x = \left(x_s^0, (x_s^n)_{n\geq 1}, (x_u^n)_{n\geq 0}\right) \in E_s \times \mathcal{E}_s \times \mathcal{E}_u$, and let $(y_s^n)_{n\geq 1}$, $(y_u^n)_{n\geq 0}$ be the sequences associated as above. If $x_s^n = x_u^n = 0$ for all $n \geq 0$, then $y_s^{n+1} = y_u^n = 0$ for all $n \geq 0$.

Then we get from (1), (2), for $n \geq 0$:

$$\kappa^{-n-1} \left| y_s^{n+1} \right| \leq \kappa^{-1} (\kappa_s + \varepsilon) \|x\|,$$
$$\kappa^{-n} |y_u^n| \leq \kappa_u^{-1} (\kappa + \varepsilon) \|x\|.$$

This shows that $(y_s^n)_{n\geq 1} \in \mathcal{E}_s$, $(y_u^n)_{n\geq 0} \in \mathcal{E}_u$ and we thus define a map $\theta \colon E_s \times \mathcal{E}_s \times \mathcal{E}_u \to \mathcal{E}_s \times \mathcal{E}_u$. Moreover, (1), (2) show that θ is Lipschitz with:

$$\operatorname{Lip} \theta \leq \max \left(\kappa^{-1} (\kappa_s + \varepsilon), \kappa_u^{-1} (\kappa + \varepsilon) \right) < 1.$$

For $x_s \in E_s$ let $\left((\varphi_s^n (x_s))_{n\geq 1}, (\varphi_u^n (x_u))_{n\geq 0} \right) = \varphi(x_s)$ be the unique fixed point of θ_{x_s}. Denote φ_u^0 by g. From Appendix A we see that φ, hence g, are contracting. From (3) we see that for any $x_s \in E_s$, $g(x_s)$ is the only vector x_u in E_u such that $(x_s, x_u) \in W_\kappa^s(f)$. This means that $W_\kappa^s(f)$ is the graph of g. We have $\varphi(0) = 0$, hence $g(0) = 0$. Let $x = (x_s, g(x_s))$, $\overline{x} = (\overline{x}_s, g(\overline{x}_s)) \in W_\kappa^s(f)$. As $f(W_\kappa^s(f)) \subset W_\kappa^s(f)$ and g is contracting, we have

$$\begin{aligned} |f(x) - f(\overline{x})| &= |f_s(x) - f_s(\overline{x})| \\ &\leq (\kappa_s + \varepsilon) |x - \overline{x}| \quad \text{(cf (1))} \end{aligned}$$

The last assertion in the theorem follows from this and $\kappa_s + \varepsilon < \kappa$. □

3.3. Differentiability of g

Theorem. *Under the assumptions of Theorem 3.1, if moreover $\kappa < 1$ and f is C^r $(1 \leq r \leq +\infty)$, then g is also C^r.*

Proof. In view of Appendix A, it is sufficient to show that θ is C^r. Let $\mathcal{E} = \{(x^n)_{n\geq 0}, x^n \in E, \sup_n \kappa^{-n} |x^n| = \left\|(x^n)_{n\geq 0}\right\| < +\infty\}$. Clearly, θ is C^r if and

only if the map $F:\mathcal{E} \to \mathcal{E}$ defined by $F\left((x^n)_{n\geq 0}\right) = (f(x^n))_{n\geq 0}$ is also C^r. Let $x = (x^n)_{n\geq 0}$, $\Delta x = (\Delta x^n)_{n\geq 0} \in \mathcal{E}$. For $n \geq 0$, $t \in [0,1]$, $1 \leq k \leq r$, define:

$$D_k(t,n) = D^k_{x^n_t} f, \qquad x^n_t = x^n + t\Delta x^n.$$

We write Taylor's formula:

$$f(x^n + \Delta x^n) = f(x^n) + \sum_{\ell=1}^{k} \frac{1}{\ell!} D_\ell(0,n)(\Delta x^n, \dots, \Delta x^n) + R_k,$$

$$R_k = \frac{1}{(k-1)!} \int_0^1 [D_k(t,n) - D_k(0,n)](\Delta x^n, \dots, \Delta x^n)(1-t)^{k-1}\, dt.$$

Because $\kappa < 1$, we have $\lim_n |x^n| = \lim |\Delta x^n| = 0$, hence $D_\ell = \sup_{t,n} |D_\ell(t,n)| < +\infty$. The ℓ-linear map

$$\left((\Delta x^n_1)_{n\geq 0}, \dots, (\Delta x^n_\ell)_{n\geq 0}\right) \mapsto (D_\ell(0,n)(\Delta x^n_1, \dots, \Delta x^n_\ell))_{n\geq 0}$$

therefore sends $\mathcal{E}^\ell$ to $\mathcal{E}$ and has norm $\leq D_\ell$. From this we conclude easily that F is C^r. $\square$

Remarks.

1. We have stated and proved the theorem for integral differentiability r. It is still valid for real $r \geq 1$ (with the same proof).
2. From the proof, one sees that $D_0 g = 0$ if $D_0 f = T$.

3.4. Stable Manifolds for Fibered Maps

Let $\pi : E \to K$ be a finite dimensional vector bundle over a compact metric space K. Let F be a linear hyperbolic automorphism of E fibered over a homeomorphism f of K. Choose $\kappa_s < 1 < \kappa_u$ such that F is $\kappa_s - \kappa_u$ hyperbolic and a continuous family $(|\ \ |_x)_{x\in K}$ of norms adapted to F.

Let $\widetilde{F} : E \to E$ be a continuous map, fibered over f. For $\kappa \in (\kappa_s, \kappa_u)$, define

$$W^s_\kappa(\widetilde{F}) = \{v \in E, \sup_{n\geq 0} \kappa^{-n} |\widetilde{F}^n(v)| < +\infty\},$$
$$W^s_\kappa(x, \widetilde{F}) = W^s_\kappa(\widetilde{F}) \cap E_x.$$

We have clearly:

$$\widetilde{F}_x(W^s_\kappa(x,\widetilde{F})) \subset W^s_\kappa(f(x), \widetilde{F}), \qquad \forall\, x \in K,$$
$$W^s_\kappa(x, F) = E^s_x, \qquad \forall\, x \in K.$$

Theorem. *Assume that $\widetilde{F}_x(0_x) = 0_{f(x)}$ for all $x \in K$, that $\widetilde{F}_x$ is Lipschitz for all $x \in K$ and that*

$$\varepsilon = \sup_K \operatorname{Lip}(\widetilde{F}_x - F_x) < \varepsilon_2 = \min(\kappa - \kappa_s, \kappa_u - \kappa).$$

Then there exists a continuous family $(g_x)_{x\in K}$ *of contracting maps* $g_x\colon E_x^s \to E_x^u$ *such that for any* $x \in K$, $W_\kappa^s(x,\widetilde{F})$ *is the graph of* g_x. *Moreover, for* $x \in K$, $v, v' \in W_\kappa^s(x,\widetilde{F})$, *we have*

$$|\widetilde{F}(v) - \widetilde{F}(v')|_{f(x)} \leq (\kappa_s + \varepsilon)\,|v - v'|_x$$
$$\lim_{n\to+\infty} \kappa^{-n}\,|\widetilde{F}^n(v)| = 0.$$

Proof. Let $\mathcal{F}\colon \Gamma_b(E) \to \Gamma_b(E)$ and $\widetilde{\mathcal{F}}\colon \Gamma_b(E) \to \Gamma_b(E)$ be the maps induced by F, $\widetilde{F}$. The linear automorphism $\mathcal{F}$ of $\Gamma_b(E)$ is $\kappa_s - \kappa_u$ hyperbolic, with associated splitting

$$\Gamma_b(E) = \Gamma_b(E^s) \oplus \Gamma_b(E^u).$$

We have $\widetilde{\mathcal{F}}(0) = 0$ and $\operatorname{Lip}(\widetilde{\mathcal{F}} - \mathcal{F}) = \varepsilon < \varepsilon_2$. Define, as in 3.1:

$$W_\kappa^s(\widetilde{\mathcal{F}}) = \{\sigma \in \Gamma_b(E), \sup_{n\geq 0} \kappa^{-n}\,|\widetilde{\mathcal{F}}^n(\sigma)| < +\infty\}$$

By Theorem 3.1, there is a contracting map $\mathcal{G}\colon \Gamma_b(E^s) \to \Gamma_b(E^u)$ whose graph is $W_\kappa^s(\widetilde{\mathcal{F}})$.

For $v \in E$, let τ_v be defined by

$$\tau_v(x) = \begin{cases} 0 & \text{if} \quad x \neq \pi(v) \\ v & \text{if} \quad x = \pi(v). \end{cases}$$

Then $\tau_v \in \Gamma_b(E)$, $|\tau_v| = |v|$ and $\widetilde{\mathcal{F}}^n(\tau_v) = \tau_{\widetilde{F}^n(v)}$ for $n \geq 0$. Consequently:

$$v \in W_\kappa^s(\widetilde{F}) \Leftrightarrow \tau_v \in W_\kappa^s(\widetilde{\mathcal{F}}).$$

Let $x \in K$, $v_s \in E_x^s$. Define $g_x(v_s) = v_u = \mathcal{G}(\tau_{v_s})(x)$, $v = v_s + v_u$. As $\tau_{v_s} + \mathcal{G}(\tau_{v_s})$ belongs to $W_\kappa^s(\widetilde{\mathcal{F}})$, v belongs to $W_\kappa^s(x,\widetilde{\mathcal{F}})$. Conversely, if $\overline{v} = v_s + \overline{v}_u$ belongs to $W_\kappa^s(x,\widetilde{F})$, then $\tau_{\overline{v}} = \tau_{v_s} + \tau_{\overline{v}_u}$ belongs to $W_\kappa^s(\widetilde{\mathcal{F}})$, hence $\tau_{\overline{v}_u} = \mathcal{G}(\tau_{v_s})$ and $\overline{v}_u = v_u$.

We have proved that $W_\kappa^s(x,\widetilde{F})$ is the graph of g_x, and g_x is contracting because $\mathcal{G}$ is. Let us show that g_x depends continuously on x. It is sufficient to show that if $\sigma_s \in \Gamma_c(E^s)$, the section σ_u defined by

$$\sigma_u(x) = g_x(\sigma_s(x))$$

is also continuous. Indeed, we can apply Theorem 3.1 to the restrictions $\mathcal{F}_c, \widetilde{\mathcal{F}}_c$ of $\mathcal{F}, \widetilde{\mathcal{F}}$ to $\Gamma_c(E)$ and get a map $\mathcal{G}_c\colon \Gamma_c(E^s) \to \Gamma_c(E^u)$ whose graph is $W_\kappa^s(\widetilde{\mathcal{F}}_c)$. Then we must have $\mathcal{G}_c(\sigma_s)(x) + \sigma_s(x) \in W_\kappa^s(x,\widetilde{F})$ for all $x \in K$, showing that $\mathcal{G}_c(\sigma_s) = \sigma_u \in \Gamma_c(E)$. The last assertions in the theorem are consequences of the similar conclusions in Theorem 3.1. $\square$

Complement. In the setting of the theorem, assume moreover that $\widetilde{F}_x$ is of class C^r for all $x \in K$, that $x \mapsto \widetilde{F}_x$ is continuous for the C^r topology, and that $\kappa < 1$. Then g_x is of class C^r for all $x \in K$, and $x \mapsto g_x$ is continuous in the C^r topology.

This is a consequence of Theorem 3.3: under the hypothesis, the map $\widetilde{\mathcal{F}}_c$ above is C^r, hence $\mathcal{G}_c$ is C^r which implies the conclusion above. Details are left to the reader.

3.5. Local Stable Manifolds

Let M be a smooth manifold, $f : M \to M$ a C^r-diffeomorphism ($r \geq 1$) and K a compact invariant hyperbolic subset.

We choose some continuous family $(|\ \ |_x)_{x\in K}$ of norms on $E = TM_{|K}$ adapted to $Tf_{|K}$. We also choose an exponential map Exp with the same properties as in 2.4.

For $\delta > 0$ sufficiently small and $x \in K$ define

$$V(x,\delta) = \exp_x(\{v \in E_x, |v|_x < \delta\}).$$

Theorem. *There exist $0 < \kappa < 1$, $\delta_1 > 0$ and a continuous family $(j_x)_{x\in K}$ of C^r embeddings:*

$$j_x\colon \{v_s \in E_x^s, |v_x^s| < \delta_1\} \hookrightarrow M$$

such that, for all $x \in K$, $0 < \delta \leq \delta_1$, the image under j_x of $\{v_s \in E_x^s, |v_x^s| < \delta\}$ is the local stable manifold of x

$$W^s(x,\delta,f) = \{y \in M,\ \forall\, n \geq 0, f^n(y) \in V(f^n(x),\delta)\}$$

Moreover we have the following properties:

(i) $x = j_x(0_x) \in W^s(x,\delta,f)$; $\ T_xW^s(x,\delta,f) = E_x^s$

(ii) $f(W^s(x,\delta,f)) \subset W^s(f(x),\kappa\delta,f)$

(iii) *let d be a distance on M associated to some Riemannian metric; there exists $c > 0$ such that for all $x \in K$, $y, y' \in W^s(x,\delta_1,f)$, $n \geq 0$, we have*

$$d(f^n(y), f^n(y')) \leq c\kappa^n d(y,y').$$

Proof. We construct $\widetilde{F}$ as in 2.4 (with $f = g$), and apply Theorem 3.4 to $\widetilde{F}$ getting a continuous family $(g_x)_{x\in K}$ of C^r contracting maps $g_x\colon E_x^s \to E_x^u$, the graph of g_x being $W_1^s(x,\widetilde{F})$. We define, for $x \in K$, $v_s \in E_x^s$, $|v_s| < \delta_1$:

$$j_x(v_s) = \exp_x(v_s + g_x(v_s)).$$

Because $\widetilde{F}_x\colon W_1^s(x,\widetilde{F}) \to W_1^s(f(x),\widetilde{F})$ is contracting, we get from property (b) of $\widetilde{F}$, as in the lemma in 3.4:

$$\exp_x^{-1}(W^s(x,\delta,f)) = W_1^s(x,\widetilde{F}) \cap \{v \in E_x, |v|_x < \delta\}$$

for all $0 < \delta \leq \delta_1$. This shows that $W^s(x,\delta,f)$ is the image under j_x of $\{v_s \in E_x^s, |v_s| < \delta\}$. Property (i) is obvious; properties (ii), (iii) follow easily from Theorem 3.4. □

Remarks.

1. Replacing f by f^{-1}, we also define local unstable manifolds of points of K, with similar properties. We observe that the *same* continuous family of norms $(|\ \ |_x)_{x\in K}$ on $TM_{|K}$, adapted to $Tf_{|K}$ (and consequently the same $V(x,\delta)$) may be used to define both the local stable and local unstable manifolds of points of K.

2. If $Tf_{|K}$ is $\kappa_s - \kappa_u$ hyperbolic for some $0 < \kappa_s < \kappa_u < 1$ we may define in a similar way the local strong stable manifolds of points of K.

3.6. Global Stable Manifolds

Let M, f, K be as above. We define the *(global) stable manifold* of a point $x \in K$ as

$$W^s(x,f) = \{y \in M, \lim_{n\to+\infty} d(f^n(x), f^n(y)) = 0\}, \tag{1}$$

where d is a distance associated to some Riemannian metric on M.

It is clear that, for any $x \in K$ we have

$$f(W^s(x,f)) = W^s(f(x), f). \tag{2}$$

From Theorem 3.5 and the definition (1) of $W^s(x,f)$ we obtain

$$W^s(x,f) = \bigcup_{n\geq 0} f^{-n}(W^s(f^n(x), \delta, f)), \tag{3}$$

for any $0 < \delta \leq \delta_1$.

From this and from Theorem 3.5 it follows that

$$W^s(x,f) = \{y \in M, \overline{\lim_{n\to+\infty}} d(f^n(x), f^n(y)) < \delta_2\}, \tag{4}$$

for some $\delta_2 > 0$ independent of x, and

$$W^s(x,f) = \{y \in M, \lim_{n\to+\infty} \kappa^{-n} d(f^n(x), f^n(y)) = 0\}, \tag{5}$$

for some $0 < \kappa < 1$ independent of x.

Finally, we have

Theorem. *The stable manifold $W^s(x,f)$ is, for any $x \in K$, the image of a C^r injective immersion $\jmath_x : E^s_x \hookrightarrow M$. The map $x \mapsto \jmath_x$ is continuous on K for the C^r topology on compact sets. The tangent space $T_x W^s(x,f)$ is E^s_x.*

Proof. Let $\widetilde{F}$ and $(g_x)_{x\in K}$ be as in the proof of Theorem 3.5. Let $x \in K$, $v \in E^s_x$. Provided n is big enough, the point

$$f^{-n} \circ \exp_{f^n(x)} \circ \widetilde{F}^n (v + g_x(v))$$

does not depend on n. Define it to be $\jmath_x(v)$. Using (3), one gets easily all the conclusions of the theorem. □

4. Basic Sets

4.1. Local Product Structure

Let f be a C^1-diffeomorphism of a smooth manifold M, d a distance on M induced by some Riemannian metric, and K a hyperbolic compact invariant set for f.

By Theorem 3.5, there exist $\delta_1 > 0$, $\theta_1 > 0$ such that, if $x, y \in K$ satisfy $d(x,y) < \theta_1^{-1}\delta$ for some $0 < \delta \leq \delta_1$, then the intersection

$$W^s(x, \delta_1, f) \cap W^u(y, \delta_1, f)$$

(notations of 3.5) consists of precisely one point, which we denote by $[x, y]$; moreover this point belongs to $W^s(x, \delta, f)$ and $W^u(y, \delta, f)$. We have $f([x, y]) = [f(x), f(y)]$ when both sides are defined.

Definition 1. The hyperbolic compact invariant set K has a *local product structure* if there exist $0 < \delta_2 < \delta_1$ such that $[x, y] \in K$ for all $x, y \in K$ with $d(x, y) < \theta_1^{-1}\delta_2$.

Definition 2. The hyperbolic compact invariant set K is *locally maximal* (or *isolated*) if there exists a neighbourhood V of K such that

$$K = \bigcap_{n \in \mathbb{Z}} f^n(V).$$

Theorem. *The following properties of K are equivalent.*

(i) *K has a local product structure;*

(ii) *K is locally maximal;*

(iii) *There exist $\delta_0 > 0$ and a neighbourhood U of K such that, if $(x_i)_{i \in \mathbb{Z}}$ is a δ_0-pseudo-orbit in U and $x \in M$ satisfies $d(f^i x, x_i) < \theta\delta_0$, then $x \in K$.*

Proof. (iii) $\Rightarrow$ (ii): Let U be a neighbourhood of K small enough to satisfy (iii) and the conclusions of the Shadowing Lemma 2.7. Let $x \in \cap_{n \in \mathbb{Z}} f^n(U)$; putting $x_i = f^i(x)$ for $i \in \mathbb{Z}$, $(x_i)_{i \in \mathbb{Z}}$ is a δ-pseudo-orbit for all $\delta > 0$. In view of the uniqueness in the Shadowing Lemma (2.7, Remark) and (iii), we have $x \in K$.

(ii) $\Rightarrow$ (i). Let V be a neighbourhood of K such that

$$\bigcap_{n \in \mathbb{Z}} f^n(V) = K$$

Let $0 < \delta_2 < \delta_1$ be such that $W^s(x, \delta_2, f)$, $W^u(x, \delta_2, f)$ are contained in V for all $x \in K$. Then, for $x, y \in K$ with $d(x, y) < \theta_1^{-1}\delta_2$, we have

$$\begin{aligned} &z = [x, y] \in W^s(x, \delta_2, f) \cap W^u(y, \delta_2, f), \\ &f^n(z) \in W^s(f^n x, \delta_2, f) \subset V \qquad \text{for } n \geq 0, \\ &f^n(z) \in W^u(f^n y, \delta_2, f) \subset V \qquad \text{for } n \leq 0, \end{aligned}$$

and hence $z \in K$.

(i) $\Rightarrow$ (iii). Let $\delta_2 > 0$ be as in the definition. With δ_0, U sufficiently small, let $(x_i)_{i \in \mathbb{Z}}$ be a δ_0-pseudo-orbit and $x \in M$ as in (iii). We may assume that $x_i \in K$ for all $i \in \mathbb{Z}$ and that x belongs to some (fixed) compact invariant hyperbolic set K' (2.7, Remark). With δ_0 sufficiently small ($\delta_0 \ll \delta_2, \delta_1'$, where δ_1' defines the local product in K'), we will show that $[x, x_0] = x^+$ and $[x_0, x] = x^-$ belong to K; then $x = [x^+, x^-] \in K$. Let $(\overline{x}_i)_{i \in \mathbb{Z}}$ be defined by $\overline{x}_i = x_i$ for $i \leq 0$ and:

$$\overline{x}_{i+1} = [x_{i+1}, f(\overline{x}_i)] \in K, \quad i \geq 0;$$

this makes sense since we have $\overline{x}_i \in W^s(x_i, c\delta_0, f)$ by induction over i for some large constant c. With δ_0 small enough, we have $f^{-1}(W^u(\overline{x}_{i+1}, \delta_1', f)) \subset W^u(\overline{x}_i, \delta_1', f)$ for $i \geq 0$, hence $f^i(x^+) = [f^i(x), \overline{x}_i]$ for $i \geq 0$ and

$$x^+ = \lim_{i \to +\infty} f^{-i}(\overline{x}_i) \in K.$$

In a similar way, one shows that $x^- \in K$. $\square$

Remark. In view of Proposition 1.5, one can always, given M, f, K as above, find a hyperbolic compact invariant set $K' \supset K$ such that $[x, y] \in K'$ for $x, y \in K$ (with $d(x, y) < \theta_1^{-1}\delta_1$); but a priori one does not have $[x, y] \in K'$ *for* $x, y \in K'$.

4.2. Spectral Decomposition

Let f be a C^1-diffeomorphism of a smooth manifold M, and K a compact invariant hyperbolic *locally maximal* subset.

We denote by $C(f_{|K})$ (resp. $P(f_{|K})$) the set of chain recurrent (resp. periodic) points of $f_{|K}$.

It follows from 2.7, Corollary and the hypothesis that K is locally maximal that we have

$$C(f_{|K}) = \overline{P(f_{|K})}.$$

Proposition. *$C(f_{|K})$ has local product structure.*

Proof. The map $(x, y) \to [x, y]$ is continuous on $\{(x, y) \in K \times K, d(x, y) < \theta_1^{-1}\delta_1\}$; it is therefore sufficient to prove that if $x, y \in P(f_{|K})$ and $d(x, y)$ is small enough, the point $z = [x, y]$ belongs to $C(f_{|K})$. Let $w = [y, x]$; as K is locally maximal, both z and w belong to K. Let p (resp. q) be a period of x (resp. y).

Let $\delta > 0$; choose $N \geq 1$ such that

$$\begin{aligned} d\left(f^{Np}(z), x\right) < \delta/2 \quad &, \quad d\left(f^{Nq}(w), y\right) < \delta/2 \\ d\left(f^{-Nq}(z), y\right) < \delta/2 \quad &, \quad d\left(f^{-Np}(w), x\right) < \delta/2. \end{aligned}$$

Let $(z_i)_{i \in \mathbb{Z}}$ be the pseudo-orbit of period $2N(p+q)$ such that:

$$\begin{aligned} z_i &= f^i(z), \quad \text{for } -Nq \leq i < Np \\ z_i &= f^{i-2Np}(w), \quad \text{for } Np \leq i < 2Np + Nq. \end{aligned}$$

It is a δ-pseudo-orbit contained in K, and $z_0 = z$. Therefore $z \in C(f_{|K})$. $\square$

We say that K is *chain recurrent* if $C(f_{|K}) = K$. Beware that a point $x \in K$ may belong to $C(f)$ but not to $C(f_{|K})$.

By the proposition above and Appendix B, the compact invariant hyperbolic subset $\widehat{K} = C(f_{|K})$ is locally maximal and chain recurrent.

Theorem (Spectral Decomposition). *Assume that K is* locally maximal *and* chain recurrent. *Then there exists a* partition *of K*

$$K = \bigcup_{i=1}^{\ell} \bigcup_{j=1}^{s_i} K_{i,j} = \bigcup_{i=1}^{\ell} K_i$$

into compact subsets, such that:

(i) $f(K_{i,j}) = K_{i,j+1}, \quad f(K_{i,s_i}) = K_{i,1}$ *for* $1 \le i \le \ell, \quad 1 \le j < s_i$;

(ii) *the restriction of* f^{s_i} *to* $K_{i,j}$ *(which is invariant under* f^{s_i} *by* (i)*) is* topologically mixing.

Proof. Let $P = P(f_{|K})$; as K is chain recurrent we have $\overline{P} = K$. Let $\mathcal{R}$ be the relation defined in P by:

$$x\mathcal{R}y \Longleftrightarrow \begin{cases} W^s(x) \cap W^u(y) \cap K \neq \emptyset \\ W^s(y) \cap W^u(x) \cap K \neq \emptyset. \end{cases}$$

To complete the proof of the theorem, we need the following lemmas.

Lemma 1. *The relation* $\mathcal{R}$ *is an equivalence relation.*

Proof of Lemma 1. $\mathcal{R}$ is clearly reflexive and symmetric. Let $x, y, z \in P$ such that $x\mathcal{R}y$, $y\mathcal{R}z$. Let $w \in W^s(x) \cap W^u(y) \cap K$, $t \in W^s(y) \cap W^u(z) \cap K$. Let p, q, r be periods of x, y, z, respectively. With N a big integer, define:

$$u_i = \begin{cases} f^{i-Npq}(w) & \text{for} \quad i \ge 0 \\ f^{i+Nqr}(t) & \text{for} \quad i < 0. \end{cases}$$

Given $\delta > 0$, this is a δ-pseudo-orbit in K if N is big enough. From Theorems 2.7 and 4.1, there exists $u \in K$ such that $d\left(f^i(u), u_i\right) < \theta\delta$ for all $i \in \mathbb{Z}$. Then we have

$$\overline{\lim_{n \to +\infty}} d\left(f^{np}(u), x\right) \le \theta\delta,$$
$$\overline{\lim_{n \to -\infty}} d\left(f^{nr}(u), z\right) \le \theta\delta.$$

With δ small enough, this implies $u \in W^s(x) \cap W^u(z)$. Similarly, we have $W^u(x) \cap W^s(z) \cap K \neq \emptyset$, and $x\mathcal{R}z$. □

Lemma 2. *The closures of the equivalence classes of* $\mathcal{R}$ *form a partition of* K *into (disjoint) open compact subsets of* K.

Proof. Indeed, K has a local product structure; if $\theta_1, \delta_2 > 0$ are as in 4.1 and $x, y \in P$ satisfy $d(x, y) < \theta_1^{-1}\delta_2$, we have $[x, y] \in K$ and $[y, x] \in K$, hence $x\mathcal{R}y$. Lemma 2 follows. □

As K is compact there are only finitely many equivalence classes. As we have, for $x, y \in P$

$$x\mathcal{R}y \Longleftrightarrow f(x)\,\mathcal{R}f(y)$$

we can label the closures of the equivalence classes as $K_{i,j}$ so that (i) is satisfied.

To prove (ii), we first observe that $L = K_{i,j}$ is locally maximal and chain recurrent for $g = f^{s_i}$. Let U, V be non empty open sets in L; choose $x \in U \cap P$, $y \in V \cap P$; let p a period of x, q a period of y (For g), $d = p \wedge q$, $0 \le i < d$. Then, we have $x\mathcal{R}g^{-i}(y)$; let $w \in W^u(x) \cap W^s\left(g^{-i}(y)\right) \cap L$; for m, m' big enough, we have $g^{-mp}(w) \in U$, $g^{m'q+i}(w) \in V$, and $g^{mp+m'q+i}(U) \cap V \neq \emptyset$; therefore, $g^{m''d+i}(U) \cap V$ for all m'' big enough, and $g_{|L}$ is topologically mixing. □

4.3. Basic Sets

Let f be a C^1-diffeomorphism of a smooth manifold M, and K a compact invariant hyperbolic *locally maximal* subset. Define the stable and unstable sets of K:

$$W^s(K) = \{y \in M, \lim_{n\to+\infty} d(f^n(y), K) = 0\}$$
$$W^u(K) = \{y \in M, \lim_{n\to-\infty} d(f^n(y), K) = 0\}.$$

Proposition. *We have*

$$W^s(K) = \bigcup_{x\in K} W^s(x,f), \quad W^u(K) = \bigcup_{x\in K} W^u(x,f).$$

Proof. Clearly we have $W^s(x,f) \subset W^s(K)$ for all $x \in K$. Let $y \in W^s(K)$, $\delta > 0$; let m be such that there exists $\overline{y} \in K$ with $d(f^m y, \overline{y}) < \delta$ and $d(f^n y, K) < \delta$ for $n \geq m$. Define a δ-pseudo-orbit $(y_i)_{i\in\mathbb{Z}}$ by

$$\begin{cases} y_i = f^i(y) & \text{if} \quad i \geq m \\ y_i = f^{i-m}(\overline{y}) & \text{if} \quad i < m. \end{cases}$$

Let $x \in M$ such that $d\left(f^i(x), y_i\right) < \theta\delta$ for all $i \in \mathbb{Z}$ (Theorem 2.7); with δ small enough, the inequalities $d\left(f^i(y), f^i(x)\right) < \theta\delta$ $(i \geq m)$ imply $y \in W^s(x,f)$, with $x \in K$ since K is locally maximal. The proof for $W^u(K)$ is similar. □

We say that K is *chain-transitive* if for all $x, x' \in K$ and all $\delta > 0$, there exist a δ-pseudo-orbit $(x_i)_{i\in\mathbb{Z}}$ and $n \geq 0$ such that $x_0 = x$, $x_n = x'$.

If K is transitive, K is chain-transitive.

If K is chain-transitive, K is chain recurrent.

Conversely, If K is a compact invariant hyperbolic set which is locally maximal and chain recurrent, we see from the spectral decomposition that K is transitive (resp. chain-transitive) if and only if there exists $s \geq 1$ and a partition in compact subsets

$$K = \bigcup_{i=0}^{s-1} f^i(L)$$

such that $f^s(L) = L$ and $f^s|_L$ is topologically mixing.

A *basic set* is a compact, invariant, hyperbolic, locally maximal, transitive subset of M; the integer $s \geq 1$ in the partition above is the *period* of the basic set.

Proposition. *Let M, f be as above and K a basic set of period 1.*

(i) *for any $x, y \in K$, $W^s(x) \cap W^u(y) \cap K$ is dense in K;*

(ii) *for any $x, y \in K$, $W^s(x) \cap W^u(y, \delta, f) \cap K$ is dense in $W^u(y, \delta, f) \cap K$, for any $\delta > 0$;*

(iii) *for any $x \in K$, $W^s(x)$ is dense in $W^s(K)$.*

Proof. We first prove that $W^s(x) \cap K$ is dense in K. Let $w \in K$.

Let $\delta > 0$; choose a point $z \in \omega(x)$. As f/K is topologically mixing, we can find $n \geq 1$ and $t \in K$ with:

$$d(t,x) < \delta, \qquad d(f^n t, z) < \delta/2, \qquad d(f^n x, z) < \delta/2.$$

Let $u \in K$ be a point satisfying

$$\begin{aligned} d(f^i u, f^i t) &< \theta\delta \qquad \text{for } i < r \\ d(f^i u, f^i x) &< \theta\delta \qquad \text{for } i \geq r. \end{aligned}$$

Then, $u \in W^s(x)$ (if δ is small enough) and $d(u,w) < (\theta+1)\delta$.

Therefore $W^s(x) \cap K$ is dense in K. Similarly $W^u(y) \cap K$ is dense in K for any $y \in K$.

Let $w \in K$; let w^+ (resp. w^-) a point in $W^s(x) \cap K$ (resp. $W^u(y) \cap K$) near w; then $w^* = [w^+, w^-] \in W^s(x) \cap W^u(y) \cap K$ is near w, proving (i).

Let $w \in W^u(y,\delta,f) \cap K$; for any $w^+ \in W^s(x) \cap K$ near w, $w^* = [w^+, w]$ belongs to $W^s(x) \cap W^u(y,\delta,f) \cap K$ and is near w; this proves (ii).

The map $z \mapsto W^s(z,\delta,f)$ is continuous in K (3.5); let

$$W^s(K,\delta) = \bigcup_{z \in K} W^s(z,\delta,f);$$

by (i), $W^s(x) \cap W^s(K,\delta)$ is dense in $W^s(K,\delta)$. As

$$W^s(K) = \bigcup_{n \geq 0} f^{-n}(W^s(K,\delta))$$

by the last proposition, we obtain (iii). □

4.4.

Our last result is the first step in the study of the *global* dynamics of hyperbolic diffeomorphisms.

Let M be a *compact* smooth manifold, and let f be a C^1 diffeomorphism of M. We denote by $P(f)$ (resp. $R(f)$, resp. $L(f)$, resp. $C(f)$) the set of periodic (resp. recurrent, resp. limit, resp. chain-recurrent) points of f in M.

The sets $\overline{P(f)}$, $\overline{R(f)}$, $\overline{L(f)}$, $\overline{C(f)}$ are invariant compact subsets of M.

Theorem. *Assume that $\overline{P(f)}$ (resp. $\overline{R(f)}$, resp. $\overline{L(f)}$, resp. $\overline{C(f)}$) is hyperbolic. Then, this set is locally maximal and chain-recurrent, hence it is a finite union of basic sets; moreover it is equal to $\overline{P(f)}$.*

Proof. Let L be the set under consideration. From Appendix B, we know that L is chain-recurrent; from 2.7, Corollary, we know that $L = \overline{P(f)}$. Let us show that, when $\overline{P(f)}$ is hyperbolic, it is locally maximal. Let U be a *compact* neighbourhood of L such that $K = \cap_{n \in \mathbb{Z}} f^{-n}(U)$ is hyperbolic. If x, y are nearby points in $P(f)$, the point $z = [x,y]$ belongs to K and is chain recurrent in K (by the proof of proposition 4.2); by 2.7, Corollary, we have $z \in \overline{P(f)}$. We conclude that $\overline{P(f)}$ has local product structure, hence it is locally maximal. □

Appendix A. Some Remarks on the Fixed Point Theorem for Contracting Maps

A.1.

Let X, Y be metric spaces, Y being complete, and $\theta: X \times Y \to Y$ be a map. We assume that there is a constant $k \in [0,1)$ such that

$$d_Y(\theta(x,y), \theta(x,y')) \leq k d_Y(y,y') \qquad \forall\, x \in X, \quad y \in Y, \quad y' \in Y.$$

Thus the map θ_x, $y \mapsto \theta(x,y)$ is contracting for all $x \in X$; we denote its unique fixed point by $\varphi(x)$. For $x \in X$, $y \in Y$, we have:

$$d_Y(y, \varphi(x)) \leq \frac{1}{1-k} d_Y(y, \theta(x,y)), \tag{1}$$

hence

$$d_Y(\varphi(x), \varphi(x')) \leq \frac{1}{1-k} d_Y(\theta(x', \varphi(x')), \theta(x, \varphi(x')), \text{ for } x, x' \in X. \tag{2}$$

A.2.

If θ is continuous, then φ is continuous. (Use (2)).

A.3.

Let $\alpha \in (0,1]$, assume that θ is uniformly Hölder in x of exponent α, i.e. that there exists $c > 0$ such that:

$$d_Y(\theta(x,y), \theta(x',y)) \leq c d_X(x,x')^\alpha, \qquad x, x' \in X, \quad y \in Y.$$

Then φ is Hölder of exponent α. (Use (2)).

A.4.

Taking $\alpha = 1$, we see that φ is Lipschitz when θ is. More specifically, if we assume that (with k as above):

$$d_Y(\theta(x,y), \theta(x',y')) \leq k \max(d_X(x,x'), d_Y(y,y')), \qquad x, x' \in X, \quad y, y' \in Y,$$

then

$$d_Y(\varphi(x), \varphi(x')) \leq k d_X(x,x'), \qquad x, x' \in X$$

(take $y = \varphi(x)$, $y' = \varphi(x')$).

A.5.

Finally, assume that X, Y are open subsets in Banach spaces E, F respectively, and that θ is C^r $(1 \leq r \leq +\infty)$.

Then φ is also C^r; indeed, for $x \in X$, we have

$$D_x \varphi = (1 - D_z^{(2)} \theta)^{-1} \circ D_z^{(1)} \theta,$$

where $z = (x, \varphi(x))$ and $D^{(1)}$, $D^{(2)}$ are the partial differentials in the x, y variables (we have $\|D^{(2)}\theta\| \leq k < 1$).

Appendix B. Recurrence in Compact Metric Spaces

We recall some of the standard definitions and basic facts about the various notions of recurrence.

Let K be a compact metric space. Recall that a *G_δ-set* (resp. F_σ-set) is a countable intersection (resp. union) of open (resp. closed) sets. By Baire's theorem, a countable intersection of G_δ-dense sets in a complete metric space is dense. Let f be a homeomorphism of K.

B.1. Periodic Points

We denote by $P_n(f)$, for $n \geq 1$, the set of fixed points of f^n; it is closed in K, invariant under f. We denote by $P(f) = \bigcup_{n\geq 1} P_n(f)$ the set of periodic points of f. It is an F_σ subset of K; $P(f)$ and $\overline{P(f)}$ are invariant under f.

B.2. Recurrent Points

A point $x \in K$ is positively (resp. negatively) *recurrent* if

$$\inf_{n\geq 1} d(x, f^n x) = 0 \qquad (\text{resp. } \inf_{n\geq 1} d(x, f^{-n}x) = 0)$$

We denote by $R^+(f)$ (resp. $R^-(f)$) the set of positively (resp. negatively) recurrent points, and by $R(f) = R^+(f) \cup R^-(f)$ the set of *recurrent* points. Both $R^+(f)$, $R^-(f)$ are G_δ subsets of K, and $R^+(f) \cap R^-(f)$ is *non empty*. One has $P(f) \subset R^+(f) \cap R^-(f)$, $f(R^+(f)) = R^+(f)$, $f(R^-(f)) = R^-(f)$.

B.3. Limit Points

Let $x \in K$; the ω-limit set (resp. α-limit set) of x is:

$$\omega(x) = \bigcap_{n\geq 0} \overline{\bigcup_{m\geq n} \{f^m(x)\}} \qquad (\text{resp. } \alpha(x) = \bigcap_{n\geq 0} \overline{\bigcup_{m\geq n} \{f^{-m}(x)\}}).$$

Both $\omega(x)$ and $\alpha(x)$ are non empty closed sets in K. One has:

$$x \in R^+(f) \Longleftrightarrow x \in \omega(x),$$
$$x \in R^-(f) \Longleftrightarrow x \in \alpha(x),$$

$$f(\omega(x)) = \omega(f(x)) = \omega(x) \quad , \quad f(\alpha(x)) = \alpha(f(x)) = \alpha(x).$$

The ω-limit set $L^+(f)$ is $\bigcup_x \omega(x)$, the α-limit set $L^-(f)$ is $\bigcup_x \alpha(x)$; the limit set $L(f)$ is $L^+(f) \cup L^-(f)$. One has

$$R^+(f) \subset L^+(f) \quad , \quad R^-(f) \subset L^-(f)$$
$$f(L^+(f)) = L^+(f) \quad , \quad f(L^-(f)) = L^-(f).$$

B.4. Non Wandering Points

A point $x \in K$ is *wandering* if it has a neighbourhood V such that $V \cap f^n(V) = \phi$ for all $n \geq 1$, *non wandering* otherwise. The non wandering set $\Omega(f)$ is the set of non wandering points. It is a closed subset of K. One has

$$\overline{L^+(f)} \subset \Omega(f) \quad , \quad \overline{L^-(f)} \subset \Omega(f)$$
$$f(\Omega(f)) = \Omega(f).$$

B.5. Chain-recurrent Points

Let $\delta > 0$; a *δ-pseudo orbit* is a sequence $(x_i)_{i\in\mathbb{Z}}$ in K such that

$$d(f(x_i), x_{i+1}) < \delta, \qquad \forall\, i \in \mathbb{Z}.$$

A *δ-pseudo orbit* $(x_i)_{i\in\mathbb{Z}}$ is *periodic* if there exists $k \geq 1$ such that $x_{i+k} = x_i$ for all $i \in \mathbb{Z}$. A point $x \in K$ is *chain-recurrent* if, for any $\delta > 0$, there exists a periodic δ-pseudo orbit $(x_i)_{i\in\mathbb{Z}}$ with $x_0 = x$. We denote by $C(f)$ the set of chain-recurrent points; it is called the *chain recurrent set*. It is a closed subset of K. One has:

$$\Omega(f) \subset C(f)$$
$$f(C(f)) = C(f).$$

Proposition. *One has:*

$$C(f/\overline{P(f)}) = \overline{P(f)}, \quad C(f/\overline{R(f)}) = \overline{R(f)}, \quad C(f/\overline{L(f)}) = \overline{L(f)},$$
$$C(f/C(f)) = C(f).$$

(But one may have $C(f/\Omega(f)) \subsetneq \Omega(f)$).

Proof. The assertion is immediate for $\overline{P(f)}$ and $\overline{R(f)}$. Let $y \in \omega(x)$ $(x, y \in K)$; for any $\delta > 0$, there exists N such that $d(f^n(x), \overline{L^+(f)}) < \delta$ for $n \geq N$. This allows us to construct δ_1-pseudo orbits $(y_i)_{i\in\mathbb{Z}}$, with $y_0 = y$, which are periodic: take $n_2 > n_1 \geq N$ such that $f^{n_1}(x), f^{n_2}(x)$ are near y, take $y_i \in \overline{L^+(f)}$ near $f^{n_1+i}(x)$ for $0 \leq i < n_2 - n_1$ and complete with periodicity $n_2 - n_1$. Therefore $\overline{L^+(f)} = C(f/\overline{L^+(f)})$. Similarly, $\overline{L^-(f)} = C(f/\overline{L^-(f)})$. The assertion for $\overline{L(f)}$ follows.

Let $x \in C(f)$; for $n \geq 0$, let $(x_i^n)_{i\in\mathbb{Z}}$ a periodic $\frac{1}{2^n}$-pseudo orbit with $x_0^n = x$. To show that $x \in C(f/C(f))$, it is sufficient to see that for any neighbourhood U of $C(f)$, all $x_i^n \in U$ if n is bigger than some n_0. But, otherwise, one could find a point $y \notin C(f)$ and sequences $(n_k)_{k\geq 0}$, $(i_k)_{k\geq 0}$ such that

$$\lim_{k\to+\infty} x_{i_k}^{n_k} = y \quad , \quad \lim_{k\to+\infty} n_k = +\infty.$$

For $k \geq 0$, consider the pseudo orbits

$$y_i^k = \begin{cases} x_{i+i_k}^{n_k} & \text{if} \quad i \notin q_k\mathbb{Z} \\ y & \text{if} \quad i \in q_k\mathbb{Z}, \end{cases}$$

where q_k is the period of $(x_i^{n_k})$. One obtains $y \in C(f)$, a contradiction. $\square$

B.6. Transitive Homeomorphisms

The homeomorphism f is *transitive* if for every non-empty open set U of K, the open set $\cup_{n\geq 0} f^n(U)$ is dense in K. Equivalently, for any non-empty open sets U, V of K, $f^n(U) \cap V$ in non-empty for some $n \geq 0$. It is sufficient to check these properties when U (resp. u, V) belong to some basis for the topology of K.

The homeomorphism f is transitive if and only if there exists $x \in K$ such that $\omega(x) = K$ (resp. $\alpha(x) = K$). If f is transitive, the set of $x \in K$ such that $\omega(x) = \alpha(x) = K$ is a G_δ-dense subset of K.

B.7. Topologically Mixing Homeomorphisms

The homeomorphism f is *topologically mixing* if for every non empty open sets U, V in K, there exists $m \geq 0$ such that $f^n(U) \cap V \neq \phi$ for all $n \geq m$. It is sufficient to check this when U, V belong to some countable basis for the topology of K.

If f is topologically mixing, then f^k is transitive for all $k \in \mathbb{Z}$, $k \neq 0$. For any $k \in \mathbb{Z}$, $k \neq 0$, f^k is topologically mixing if and only if f is.

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